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# The human reward system encodes the subjective value of ideas during creative thinking

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Creative thinking involves the evaluation of one's ideas in order to select the best one, but the cognitive and neural mechanisms underlying this evaluation remain unclear. Using a combination of creativity and rating tasks, this study demonstrates that individuals attribute subjective values to their ideas, as a relative balance of their originality and adequacy. This relative balance depends on individual preferences and predicts individuals' creative abilities. Using functional Magnetic Resonance Imaging, we find that the Default Mode and the Executive Control Networks respectively encode the originality and adequacy of ideas, and that the human reward system encodes their subjective value. Interestingly, the relative functional connectivity of the Default Mode and Executive Control Networks with the human reward system correlates with the relative balance of adequacy and originality in individuals' preferences. These results add valuation to the incomplete behavioral and neural accounts of creativity, offering perspectives on the influence of individual preferences on creative abilities.

From global issues to daily struggles, humans consistently face new challenges and must rely on their creativity to solve them. In neuroscience, creativity is consensually defined as the ability to form ideas that are both original and adequate[1–4]. Original ideas are unusual, novel or unique, and adequate ideas are appropriate, efficient, and useful. Despite the definition of creativity, it remains unclear whether originality and adequacy are assessed during the creative process, and whether and how this guides the production of ideas. In parallel, the fact that two people rarely arrive at the same solution to a problem and do not prefer the same idea when asked for the best alternative[5–7] suggests the critical role of subjective preferences in creativity, a topic that has also been largely overlooked in creativity research.

The current research builds on the dominant neurocognitive view for creativity, which describes it as a dual process, divided into a generation phase and an evaluation phase[8–15]. In the generation phase, individuals form novel associations: they link seemingly unrelated concepts and produce a wide variety of potentially creative ideas. This phase can be facilitated by memory structure[16–19], associative abilities[10,18,20,21] and cognitive flexibility[22,23], enabling individuals to navigate between different concepts during idea generation. The generation phase has been the main focus of creativity research in the past.

On the other hand, the evaluation phase of the creative process, during which candidate ideas are assessed for their creativity, has been largely overlooked. Yet, this step is critical, as shown in studies where individuals with better evaluation skills achieved greater creative performance[5,7,24–26] (see Guo et al.[27] for a meta-analysis). The existing findings describe creative evaluation as a deliberate and goal-directed process that relies on cognitive control[13,21,28] and memory-search processes[29], whose accuracy highly depends on context and personality[30].

However, while creativity is consensually defined by originality and adequacy, it is still unknown whether these two dimensions are assessed during creative evaluation and whether this guides the production of ideas. Some studies have suggested that adequacy might be less important in the evaluation of the creativity of an idea[31,32], while others proposed that it depends on individuals' personality or the context of the creative task[6,7,33]. Another recent study provides evidence for the interaction of the dimensions during idea evaluation[34]. Aside from these few insights, the integration of originality and adequacy and the exact computations at play during the evaluation step of creativity remain undetermined.

In this study, we conceptualize the evaluation step of creativity as a specific case of decision-making: similar to food options in dietary choices, creative options are compared, and the most satisfactory one is selected[35,36].

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We hypothesize that creative evaluation involves individual preferences and decision-making computations that methods from neuroeconomics can enable us to study. Specifically, we argue that creative evaluation involves a valuation process.

In neuroeconomics, valuation consists of assigning a subjective value to an item, reflecting how valuable the item is, how much the subject likes it. This subjective value then guides selection[37,38]. Similarly, we propose that the subjective value of a creative idea reflects the satisfaction it provides and plays a crucial role in guiding idea selection. In a previous work[36], we demonstrated that creativity indeed involves valuation. In particular, that study first showed that the speed at which individuals produced ideas correlated to how much they liked them, suggesting that subjective valuation of ideas energizes creative production. Second, the value given to an idea was a function of its originality and adequacy, indicating that subjective valuation integrates the fundamental criteria of creativity. This integration varied across individuals, with some giving more weight to originality than adequacy. Finally, we showed that individuals favoring originality or a balance of originality and adequacy yielded greater creativity (based on word-association performance, alternative uses fluency, and self-reports) than those favoring adequacy, showing the impact of valuation patterns on creative abilities. Overall, this seminal study empirically established that originality and adequacy are integrated into the valuation of ideas, and that valuation plays a central role in creativity. In other words, the dimensions defining a creative product (originality and adequacy) are implemented in the creative process (originality and adequacy are used to form subjective values, which guide idea production). In the current study, we replicate those results and identify the neural underpinnings of the evaluation processes involved in creative thinking.

At the neural level, creativity research has found the generation and evaluation phases to be associated with the Default Mode Network (DMN) and the Executive Control Network (ECN), respectively[28,39,40]. The DMN is a network composed of regions of the medial prefrontal cortex (mPFC), the posterior cingulate cortex (PCC), the precuneus, the inferior parietal lobule (IPL), and the temporal lobe. Its activity is observed in individuals at rest, unfocused on their environment[41]. The DMN also has a role in essential cognitive functions, including associative thinking, daydreaming, mind-wandering, and self-referential thoughts[42]. In parallel, creativity research has found the evaluation phase of the dual process to be associated with the ECN. The ECN includes regions of the dorsolateral prefrontal cortex (dlPFC), the anterior cingulate cortex (ACC), the inferior parietal lobule (IPL), and the temporal lobe. It has been consistently associated with cognitive control, working memory processes, goal-directed behavior, task switching, and decision-making[43]. Interestingly, the DMN and ECN typically exhibit mutually exclusive activity depending on whether the task is internally or externally focused[44]. On the contrary, in the context of creativity, their cooperation has been widely observed and appears to be essential for creative thinking and performance[28,40,45–47].

Despite the growing interest in creative evaluation, the underlying decision-making processes and their neural substrates have been overlooked, and a notable blind spot has emerged in our understanding of the neurocognitive bases of creative evaluation.

In neuroeconomics, research on the neural bases of decision-making processes shows that subjective values are encoded in the human reward system, also known as the brain valuation system (BVS)[48,49]. The BVS overlaps the reward system identified in animal literature[50]. It comprises the ventromedial prefrontal cortex (vmPFC), the orbitofrontal cortex (OFC), and subcortical regions, particularly the ventral striatum (VS) and the ventral tegmental area (VTA). The valuation process in the BVS is both generic and automatic. Generic in the sense that the activity in the BVS encodes the value of an object regardless of its nature. Automatic in that the value is encoded in the BVS even when an individual is evaluating another attribute, such as the age of a painting[49,51–53]. Besides the BVS, the ECN is also engaged in decision-making processes: it ensures top-down control by adjusting the values of options based on the context of the decision[54]. The ECN is also responsible for integrating the difference in values between options, which drives the selection process[55].

Based on its generic properties and its cooperation with the ECN, we hypothesize that idea valuation during creativity is supported by the BVS. This is consistent with several creativity studies, demonstrating the involvement of BVS regions[56–58] or dopaminergic modulation[59–62] in creativity. Despite these findings and the overall recognized importance of motivation in creativity, the probable involvement of the BVS in creativity has been largely overlooked.

In summary, we hypothesize in the current study that, within the creative process, idea evaluation can be subdivided into the monitoring of the idea (originality and adequacy assessment) and the valuation of the idea (subjective value assignment). We propose to investigate the cognitive mechanisms and the interactions of these processes. Notably, using functional Magnetic Resonance Imaging (fMRI), we aim to clarify whether and where originality, adequacy and subjective values are encoded during the creative process.

For this purpose, we recruited forty participants who performed a creative word association task during an fMRI session. Participants also rated how likeable, original and adequate the associations were. We first replicated the behavioral results of our previous work[36], showing that valuation depends on originality and adequacy and plays a role in creativity. Then, we defined functional localizers by identifying the brain regions encoding ideas' originality, adequacy and subjective values during the rating task. Based on these functional localizers, we demonstrated that these three dimensions are encoded in the brain when producing creative ideas in the word association task.

## Results

The study consisted of several successive tasks (Fig. 1): a Free Generation of Associates Task (FGAT), a likeability rating task and a choice task, all performed in an MRI, followed by an originality and adequacy rating task and a battery of creativity tests completed outside of the MRI. Thirty-eight subjects were included in the analyses (19 females, age = 26.5 ± 0.7 (M ± SEM), years of education = 16.7 ± 0.4 (M ± SEM)).

### Behavioral results

The Free Generation of Associates Task (Fig. 1B) is a remote thinking task reported to capture critical aspects of creativity[63,64]. In the FGAT-first condition, considered a control condition, participants were presented with a cue word and had to give the first word that came to mind when reading it. In the FGAT-distant condition, they had to give a word that was farther from the cue word, with the instruction to think creatively (see Supplementary Methods for the detailed instructions). Then, participants rated how much they liked the two-word associations and how original and adequate they found them on a pseudo-continuous scale going from 'not at all' to 'very much' (Fig. 1C, see Methods for more details). The following analyses refer to these ratings as likeability ratings (or subjective values), originality ratings and adequacy ratings.

### Individuals achieve higher adequacy and originality when instructed to be creative

To characterize participants' creative productions, we first explored the differences between FGAT "first" and "distant" responses, using the participants' adequacy and originality ratings. As expected, participants rated "first" associations as highly adequate and little original (adequacy rating = 83.1 ± 1.4; originality rating = 36.8 ± 1.5, M ± SEM across participants), while they found "distant" associations to be highly adequate and original (adequacy rating = 73.1 ± 1.4; originality rating = 63.2 ± 1.3, M ± SEM across participants).

In fact, participants found that "distant" associations were far more original than "first" associations (mean difference in originality ratings of "distant"-"first" = 26.4 ± 1.6, M ± SEM; paired two-tailed t-test: $t(37) = 16.3$, $p < 0.001$). On the other hand, they found that "distant" associations were slightly less adequate than "first" associations (mean difference in adequacy

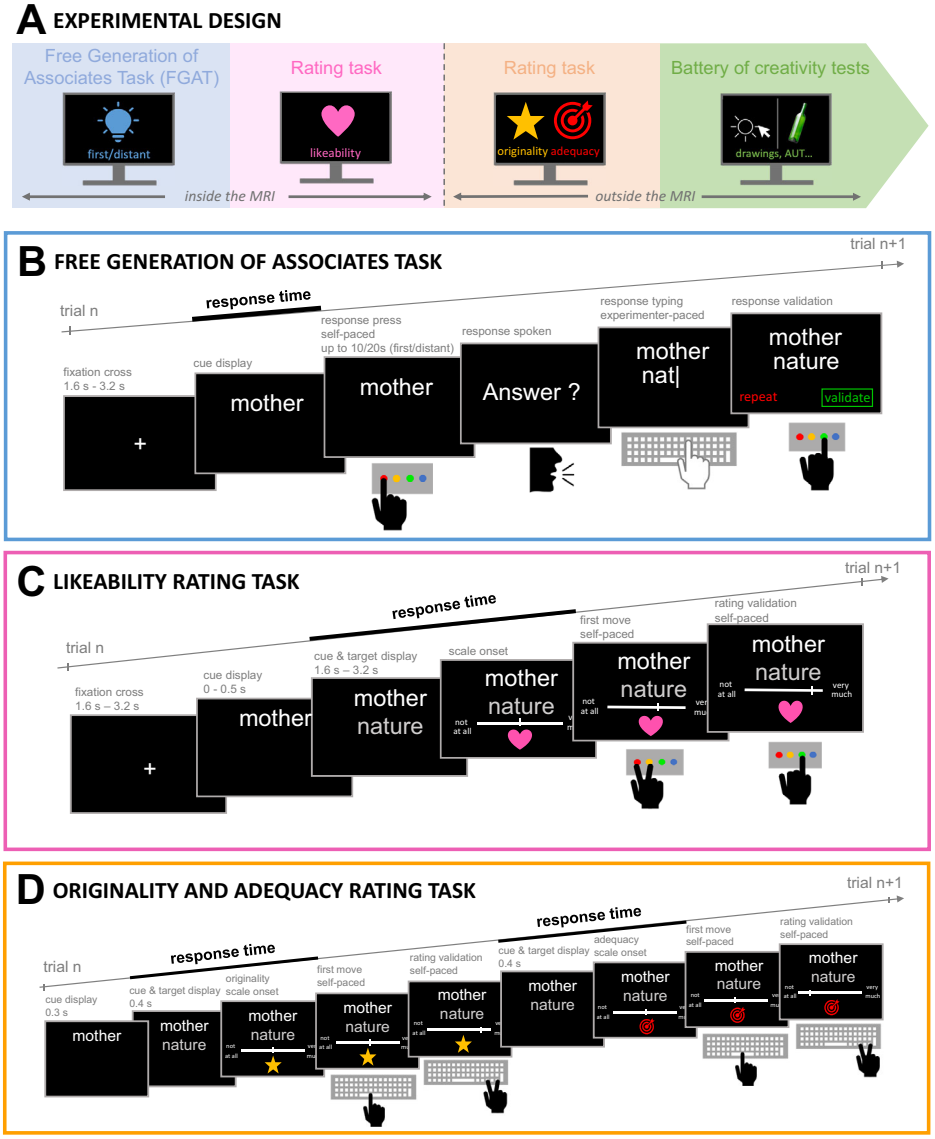


**Fig. 1 | Experimental design. A** Overview of the different tasks in chronological order. **B** Free Generation of Associates Task (FGAT). In two conditions, participants had to provide: the first word that came to their mind when they saw the cue word (FGAT-first) or a more distant word, with the instruction to be creative (FGAT-distant). Participants in the MRI provided their responses orally to the experimenter, who typed them. They had the opportunity to repeat their response in case the experimenter had misheard. **C** Likeability rating task. Participants rated each association on a likeability scale from "not at all" to "very much" and validated the rating using MRI response buttons. **D** Originality and adequacy rating task. Participants rated each association on a scale going from "not at all" to "very much" using the keyboard arrows and validated the rating using the spacebar. In the MRI, all trials of all tasks began with a fixation cross, followed by the trial screen. After the likeability rating task, participants also completed a choice task in the MRI scanner, for which data was not analyzed for the current study.

ratings of "distant"-"first" = -10.1 ± 1.4; paired two-tailed t-test: t(37) = -7.0, $p < 0.001$). Critically, the difference in originality between "first" and "distant" associations was greater than the difference in adequacy (paired two-tailed t-test: t(37) = 10.4, $p < 0.001$), indicating that "distant" associations were both original and adequate - i.e., creative - while "first" associations were mainly adequate (Fig. 2A left).

On top of the participants' self-ratings, we computed an objective measure of the rarity of the responses using the dictionary of verbal associations "Dictaverf"[65]. We found that "distant" responses were less frequent than "first" responses (paired two-tailed t-test: t(37) = -23.7, $p < 0.001$), confirming that the "distant" condition elicited more remote ideas, a measure historically associated with verbal creativity[66].

### Subjective values predict individuals' response times

Next, we investigated whether valuation plays a role during creative idea production. We looked for a signature of valuation in the FGAT-distant condition, in the shape of a relationship between valuation variables (likeability ratings) and creative production variables (FGAT-distant response times, defined as the time between cue onset and response button-press, Fig. 1B). Indeed, if likeability ratings predict FGAT-distant response times, i.e., if individuals produce the ideas they like faster, it would suggest that valuation is active during idea production.

We found that, in the FGAT-distant condition, the more the participants liked an idea, the faster they produced it ($RT_{distant}$ regressed against likeability: $\beta_{distant} = -0.18 \pm 0.02$, M ± SEM; one-sample two-tailed t-test: t(37) = -7.48, $p < 0.001$). This was not significant in the FGAT-first condition ($RT_{first}$ regressed against likeability: $\beta_{first} = 0.04 \pm 0.04$, M ± SEM; one-sample two-tailed t-test: t(37) = 1.26, $p = 0.2$). This difference between conditions was significant (effect of likeability on "distant" versus "first" RT - i.e. $\beta_{distant}$-$\beta_{first}$: paired two-tailed t-test: t(37) = -4.74, $p < 0.001$, Fig. 2A right). Control analyses confirmed that FGAT-distant response times were still predicted by likeability ratings even after adding square likeability (a proxy for confidence) as a regressor[67,68] (Supplementary Results).

Overall, this result suggests the presence of an implicit valuation process during the production of creative associations (FGAT-distant), which is absent in the production of common associations (FGAT-first). In other words, when producing creative ideas, participants intrinsically value them: this is reflected in their response time when they provide their preferred ideas faster.

### Individuals' subjective values depend on originality and adequacy

Next, we investigated how likeability ratings integrated adequacy and originality ratings. Using the data from the likeability rating task and the

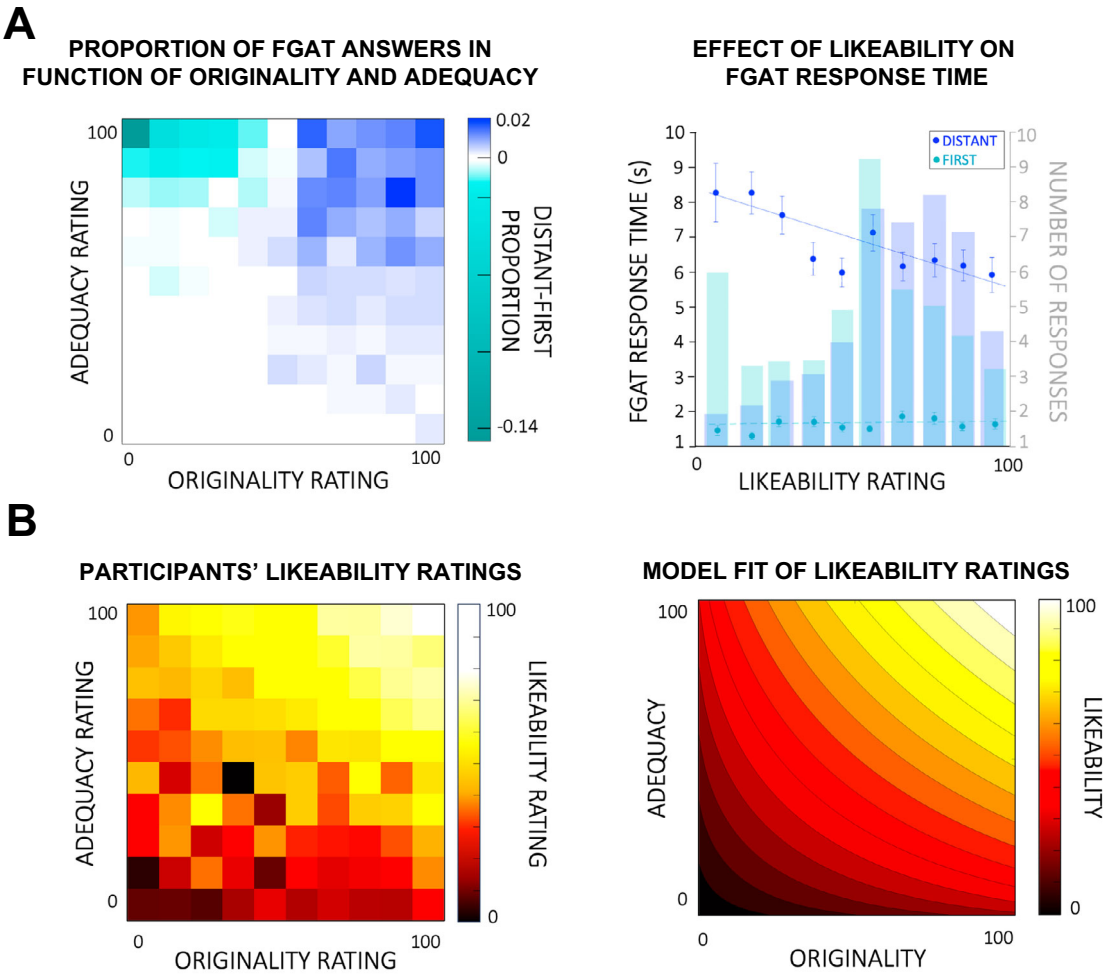


**Fig. 2 | Behavioral results of creative idea production and evaluation. A** Creative idea production. Left: Heatmap of the difference in proportion between "distant" and "first" associations per bin of adequacy and originality ratings. Bins with positive values count more "distant" answers than "first" answers and vice versa. Right: Correlation between response time and likeability ratings of the FGAT responses for the "first" and "distant" conditions. Background bars indicate the mean number of answers per bin of likeability ratings. Dots are bins of averaged participant data. Error bars are intersubject standard errors of the mean (SEM). Lines correspond to the linear regression fit at the group level in the "distant" condition (significant fit, $p < 0.05$) and the "first" condition (non-significant fit, $p > 0.05$). **B** Creative idea evaluation. Left: Heatmap of participants' average likeability ratings per bin of adequacy and originality ratings. Right: CES model fit of likeability ratings in function of adequacy and originality ratings. The verticality of isolines is captured by the α parameter (preference for originality), while the convexity of isolines is captured by the δ parameter (preference for a trade-off of both dimensions rather than extremes). $n = 38$.

originality and adequacy rating task, we observed that likeability increased with adequacy and originality: the more original and adequate an association was, the more it pleased participants (Fig. 2B left).

To quantify this relationship, we used the Constant Elasticity of Substitution (CES) model[69,70] (see Methods for the model equation). This model was used in our previous study[36], where it provided the best fit for similar data in a model comparison of twelve different models. We confirmed that the CES fitted the present dataset ($r^2 = 0.42 \pm 0.03$, M ± SEM; one-sample two-tailed t-test: $t(37) = 15.43$, $p < 0.001$; Fig. 2B right).

The CES model has two free parameters. First, the alpha parameter ($\alpha$) captures the weight given to originality relative to adequacy in the likeability ratings. In the CES equation (that you can find in the Methods), if α is greater than 0.5, likeability ratings are more driven by originality ratings than adequacy ratings, which means that the participant puts more weight on originality than adequacy (and vice versa). Second, the delta parameter (δ) captures the preference for an equilibrium of originality and adequacy. In the CES equation, if δ is lower than 1, likeability ratings increase with both adequacy and originality ratings, which means that the participant prefers an equilibrium of the two dimensions. If δ is greater than 1, likeability ratings increase when adequacy ratings are higher than originality ratings (or vice versa), which means that the participant prefers extreme adequacy and extreme originality over equilibriums.

For each participant, we estimated the model's free parameters (hereafter referred to as valuation parameters). At the group level, we found that participants' likeability ratings resulted from a balanced trade-off of originality and adequacy (no preference for originality: $\alpha = 0.46 \pm 0.04$, M ± SEM; one-sample two-tailed t-test against 0.5: $t(37) = -0.92$, $p = 0.36$; preference for equilibrium: $\delta = 0.44 \pm 0.21$, M ± SEM; one-sample two-tailed t-test against 1: $t(37) = -2.67$, $p = 0.011$). In other words, associations that were judged both original and adequate (rather than very original or very adequate) yielded higher likeability ratings at the group level. This suggests that valuation similarly and conjointly considers an idea's originality and adequacy.

Further analyses looking at the relationship between likeability ratings and response frequency are reported in the Supplementary Results.

### Individuals' subjective valuation predicts creative abilities

Then, we tested how participants' valuation parameters related to their creative abilities, as measured by a battery of tests and questionnaires. This

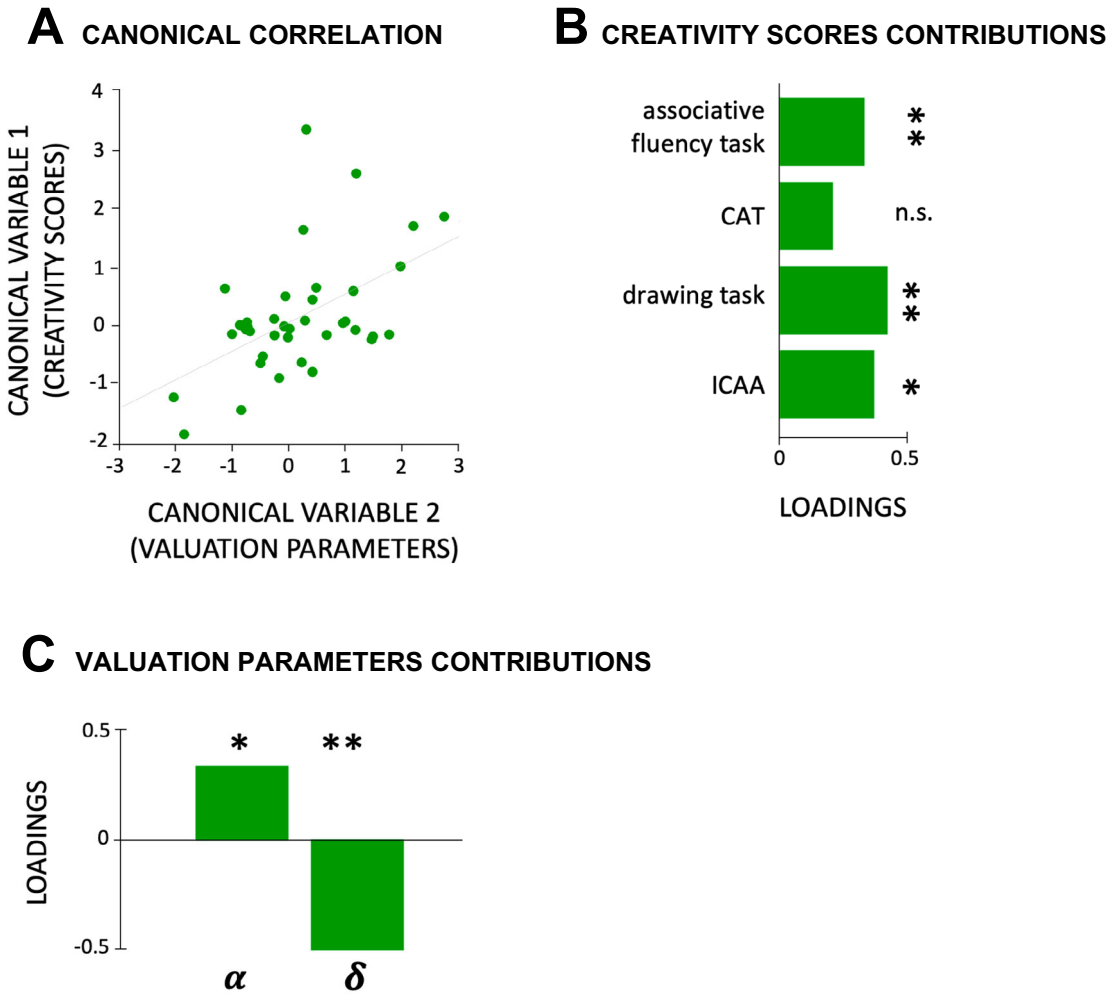


**Fig. 3 | Canonical correlation between valuation parameters and creative abilities. A** Canonical correlation between scores in a battery of creativity tests and valuation parameters ($\alpha$ and $\delta$) from the CES model of the participants' ratings. Each circle represents one participant. The line represents a significant correlation ($p < 0.05$). **B** Contribution of the creativity scores from the battery of tests. **C** Contribution of the valuation parameters. * and ** indicate statistical significance (respectively $p < 0.05$ and $p < 0.01$). n.s. stands for not significant ($p > 0.05$). $n = 38$.

battery measured associative abilities, divergent thinking and convergent thinking skills (see Methods for more details on how creativity scores were computed for each task).

We used a canonical correlation analysis and found one canonical variable showing significant dependence between the set of valuation parameters and the set of creative ability measures ($r = 0.52$, $p = 0.025$, Fig. 3A). All but one of the creativity scores significantly contributed to the canonical variable (associative fluency task score: $r = 0.34$, $p = 0.035$; CAT score: $r = 0.21$, $p = 0.20$; drawing task score: $r = 0.44$, $p = 0.006$; ICAA score: $r = 0.38$, $p = 0.018$, Fig. 3B). Both valuation parameters significantly contributed to the canonical variable ($\alpha$: $r = 0.34$, $p = 0.04$; and $\delta$: $r = -0.50$, $p = 0.0013$, Fig. 3C).

These results indicate that individuals' valuation pattern is related to their creative abilities: participants who valued originality (i.e., had a high $\alpha$ parameter) and its balance with adequacy (i.e., had a low $\delta$ parameter) yielded higher scores in the battery of creativity tests.

### Neuroimaging results

Behavioral results indicate that the production of creative ideas involves a subjective valuation process that integrates the originality and adequacy of the ideas. Then, we wanted to test the hypothesis that if likeability, originality, and adequacy were actually considered by participants, then they should be represented in the brain during the production of creative ideas. For this purpose, we explored the neural representation of (1) likeability and (2) originality and adequacy during idea production.

For each of these explorations, we used the following approach: (i) we used the likeability rating task (which isolates evaluation processes) as a localizer to locate the brain regions encoding each dimension during idea evaluation; (ii) we precisely characterized these brain regions to find the brain network they belonged to; and (iii) we tested whether the brain regions identified during idea evaluation (the likeability rating task) also encoded their respective dimension during creative idea production (the FGAT-distant).

Finally, we tested whether the connectivity pattern between the three identified networks was related to the participants' preferences (i.e., valuation parameters) identified at the behavioral level.

### Likeability is encoded in the BVS during creative idea production

Locating brain regions encoding likeability (results from the likeability rating task): We used a whole-brain parametric modulation approach to investigate the neural representation of likeability during the likeability rating task, with likeability ratings as regressors. The generalized linear model (GLM) we tested contained the likeability of the responses as a parametric modulator of a categorical boxcar regressor between the onset of the cue (i.e., the two-word association) and the first button-press to provide the likeability rating. We found a significant correlation between the associations' likeability ratings and the blood-oxygen-level-dependent (BOLD)

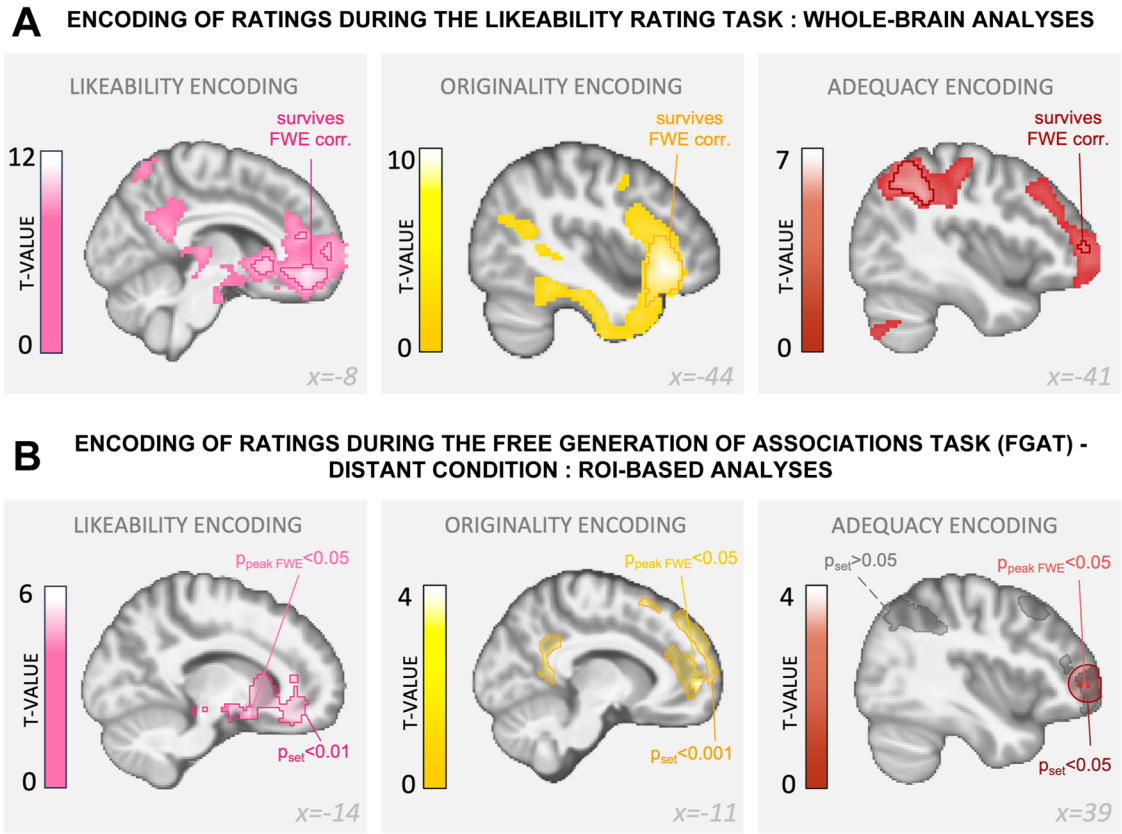


**Fig. 4 | Neuroimaging results of likeability, originality and adequacy encoding during idea evaluation and creative idea production. A** Whole-brain analysis of the neural encoding of likeability ratings (pink), originality ratings (yellow) and adequacy ratings (red) during the likeability rating task. The color code indicates the T-value of one-sample t-tests, $p < 0.001$ uncorrected. Circled regions survived cluster-level FWE correction ($p < 0.05$). See Table 1 for all significant clusters. **B** ROI-based analysis of the neural encoding of likeability ratings (pink), originality ratings (yellow) and adequacy ratings (red) during the FGAT-distant. Circled regions are the ROIs (the overlap of the regions in A and their most similar atlas network, as depicted in Fig. 5). They were used as inclusive masks for small volume correction, to identify significant peaks with FWE correction ($p < 0.05$). Colored circled regions are significant at the set level ($p < 0.05$); grey circled regions are not significant at the set level ($p > 0.05$). The color code of the peaks within the masks indicates the T-value of one-sample t-tests, displayed at $p < 0.001$ uncorrected, but surviving FWE correction ($p < 0.05$) within the mask. See Table 2 for more information. $n = 38$.

signal in regions classically identified as parts of the BVS (Fig. 4A): the vmPFC (peak Montreal Neurological Institute (MNI) coordinates: [x = -8 y = 48 z = -10], t(37) = 8.20, $p$FWE = $5.10^{-15}$) and the ventral striatum of the left ([-14, 22, 0], t(37) = 7.55, pFWE = $1.10^{-7}$) and right hemisphere ([12, 22, -2], t(37) = 6.98, $p$FWE = $4.10^{-6}$, see Table 1 for all significant clusters). This localizer supports our hypothesis that likeability ratings are encoded in the BVS during idea valuation.

Identifying the BVS: To confirm that likeability ratings involved the BVS, we compared our localizer with the BVS (as defined by the term-based meta-analyses of the Neurosynth platform https://neurosynth.org/analyses/terms/reward/), as well as 7 intrinsic functional networks from the Yeo atlas[71]. For this comparison, we quantified the number of voxels in common between our localizer and each atlas network (Fig. 5). We found that the regions reflecting likeability ratings mostly overlapped with the BVS network (31 688 voxels out of 112 370, i.e. 28% of overlap), more so than with the other 7 networks, which aligned with our observations and hypotheses. For the following analysis, we defined the BVS region of interest (ROI, the dark pink areas in Fig. 5A) as the overlapping voxels of the localizer (the light pink areas in Fig. 5A) and the BVS Neurosynth network (the black areas in Fig. 5A).

Encoding of the likeability in the BVS during creative idea production: We performed ROI-based parametric modulation analyses on the FGAT-distant to determine whether likeability was also encoded in the BVS during the production of creative ideas and specify which regions in particular. The GLM we tested contained the likeability of the responses (obtained in the rating task) as a parametric modulator of a categorical boxcar regressor between the onset of the cue and the button press to provide the creative response during the FGAT-distant.

We found a significant correlation with likeability within the BVS ROI (set $p$ value = 0.007; see Table 2 for details). Plus, when investigating the clusters within the ROI, we found a significant correlation with likeability in the ventral striatum ([-14, 8, -14], t(37) = 4.64, peak p-value (FWE) = 0.015; Fig. 4B), which confirms our hypothesis that the BVS encodes the subjective value of ideas during creative idea production.

Note that, since likeability ratings correlated with response times during the FGAT-distant, we compared the present result (the parametric modulation of the FGAT-distant BOLD signal by likeability ratings) to the parametric modulation of the FGAT-distant BOLD signal by response times. This control analysis revealed no significant correlation between response times and the BOLD signal in regions of the BVS, suggesting that the present results are not confounded with RT and might be specific to the encoding of valuation (Supplementary Results, Figure S2, and Table S2).

Interestingly, the same GLM did not yield any significant results when applied to the FGAT-first condition (peak $p$ values (FWE) > 0.5), regardless of whether we used the ROI as an inclusive mask or not. This suggests that valuation was only at play during the creative idea production (FGAT-distant condition), as opposed to when subjects formed spontaneous associations (FGAT-first condition).

Together, these analyses confirm our central hypothesis that, during creative idea production, the evaluation phase partly relies on the assignment of subjective values encoded in the BVS. In the following sections, we

## Table 1 | fMRI results of the likeability rating task

| parametric modulator: likeability rating | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **cluster label** | **cluster size** | **peak side** | **x** | **y** | **z** | **Z-score** | **T-value** | **cluster p-value** | **network of interest** |
| occipital cortex | 804 | R | 6 | -82 | -2 | >8 | 12.81 | <10–16 | none |
| **ventromedial prefrontal cortex** | **324** | **L** | **-8** | **48** | **-10** | **6.15** | **8.20** | **5.10–15** | **DMN** |
| **ventral striatum** | **91** | **L** | **-14** | **22** | **0** | **5.84** | **7.55** | **1.10–7** | **BVS** |
| **ventral striatum** | **57** | **R** | **12** | **22** | **-2** | **5.54** | **6.98** | **4.106** | **BVS** |
| cerebellum | 30 | R | 12 | -52 | -50 | 5.36 | 6.65 | 1.10–4 | none |
| anterior cingulate cortex | 18 | L | -11 | 42 | 16 | 5.33 | 6.60 | 6.10–4 | DMN |
| medial frontal pole | 13 | L | -8 | 60 | 6 | 4.94 | 5.92 | 0.002 | DMN |
| midbrain | 1 | L | -6 | -18 | -24 | 4.89 | 5.85 | 0.027 | BVS |
| **parametric modulator: originality rating** | | | | | | | | | |
| **cluster label** | **cluster size** | **peak side** | **x** | **y** | **z** | **Z-score** | **T-value** | **cluster p-value** | **network of interest** |
| **inferior frontal gyrus** | **1163** | **L** | **-44** | **28** | **-4** | **7.44** | **11.46** | **<10–16** | **none** |
| **middle temporal gyrus** | **543** | **L** | **-56** | **-15** | **-12** | **6.64** | **9.30** | **<10–16** | **DMN** |
| **dorsomedial prefrontal cortex** | **320** | **L** | **-6** | **48** | **16** | **6.55** | **9.10** | **3.10–15** | **DMN** |
| **lateral orbitofrontal cortex** | **105** | **R** | **39** | **30** | **-10** | **5.85** | **7.57** | **3.10–8** | **BVS** |
| **pre-supplementary motor area** | **72** | **L** | **-4** | **20** | **58** | **5.80** | **7.48** | **6.10–7** | **DMN** |
| ventral tegmental area | 27 | L | -4 | -22 | -20 | 5.78 | 7.43 | 1.10–4 | BVS |
| lateral orbitofrontal cortex | 35 | R | 26 | 18 | -22 | 5.63 | 7.16 | 5.10–5 | none |
| medial frontal pole | 31 | L | -4 | 62 | -14 | 5.54 | 6.99 | 8.10–5 | BVS |
| occipital cortex | 49 | L | -18 | -95 | -7 | 5.54 | 6.99 | 8.10–6 | none |
| cerebellum | 31 | R | 2 | -55 | -42 | 5.49 | 6.89 | 8.10–5 | none |
| brainstem | 150 | L | -4 | -30 | -2 | 5.47 | 6.84 | 5.10–10 | BVS |
| occipital cortex | 87 | R | 24 | -90 | -4 | 5.38 | 6.68 | 1.10–7 | none |
| cerebellum | 39 | R | 19 | -78 | -40 | 5.37 | 6.66 | 3.10–5 | none |
| fusiform gyrus | 10 | L | -34 | -8 | -42 | 5.15 | 6.29 | 0.002 | none |
| amygdala | 7 | L | -16 | -2 | -17 | 4.95 | 5.95 | 0.005 | BVS |
| temporal pole | 3 | R | 42 | 15 | -27 | 4.91 | 5.88 | 0.013 | none |
| fusiform gyrus | 2 | L | -38 | -10 | -37 | 4.85 | 5.78 | 0.018 | none |
| posterior cingulate cortex | 3 | L | -8 | -50 | 30 | 4.82 | 5.74 | 0.013 | DMN |
| globus pallidus | 3 | R | 14 | 0 | -14 | 4.82 | 5.73 | 0.013 | BVS |
| inferior frontal gyrus | 1 | L | -48 | 18 | 20 | 4.77 | 5.65 | 0.026 | ECN |
| cerebellum | 1 | R | 22 | -40 | -47 | 4.75 | 5.63 | 0.026 | none |
| **parametric modulator: adequacy rating** | | | | | | | | | |
| **cluster label** | **cluster size** | **peak side** | **x** | **y** | **z** | **Z-score** | **T-value** | **cluster p-value** | **network of interest** |
| **precuneus** | **192** | **L** | **-6** | **-70** | **38** | **5.96** | **7.80** | **3.10-11** | **ECN** |
| **intraparietal lobule** | **398** | **L** | **-41** | **-48** | **38** | **5.95** | **7.77** | **1.10-16** | **ECN** |
| **posterior cingulate cortex** | **129** | **L** | **-6** | **-25** | **30** | **5.86** | **7.59** | **5.10-9** | **ECN & BVS** |
| **lateral frontal pole** | **110** | **L** | **-36** | **52** | **3** | **5.74** | **7.36** | **4.10-6** | **ECN** |
| **intraparietal lobule** | **122** | **R** | **44** | **-38** | **43** | **5.57** | **7,04** | **9.10-9** | **ECN** |
| **inferior temporal sulcus** | **34** | **L** | **-61** | **-30** | **-17** | **5.28** | **6.50** | **7.10-5** | **DMN** |
| cerebellum | 15 | R | 44 | -65 | -50 | 5.23 | 6.41 | 0.001 | none |
| **lateral frontal pole** | **55** | **R** | **34** | **62** | **6** | **5.12** | **6.24** | **5.10-6** | **ECN** |
| lingual gyrus | 10 | R | 6 | -78 | -7 | 05.08 | 6.17 | 0.003 | none |
| lingual gyrus | 10 | R | 16 | -72 | -12 | 05.04 | 6.10 | 0.003 | none |
| medial frontal pole | 9 | L | -8 | 75 | 3 | 4.94 | 5.93 | 0.003 | none |
| supramarginal gyrus | 2 | L | -51 | -35 | 43 | 4.78 | 5.66 | 0.019 | ECN |
| lateral frontal pole | 2 | R | 42 | 62 | -12 | 4.76 | 5.64 | 0.019 | none |

The clusters listed here result from whole-brain parametric modulation analyses and survived cluster-level FWE correction ($p < 0.05$). Coordinates x, y and z refer to the Montreal Neurological Institute space. The last column indicates whether the location of the peak activation falls within a network of interest. DMN and ECN are defined by Yeo et al.'s atlas[71], and BVS is defined by the term-based meta-analyses of the Neurosynth platform https://neurosynth.org/analyses/terms/reward/). Only the results in bold are reported in the text.

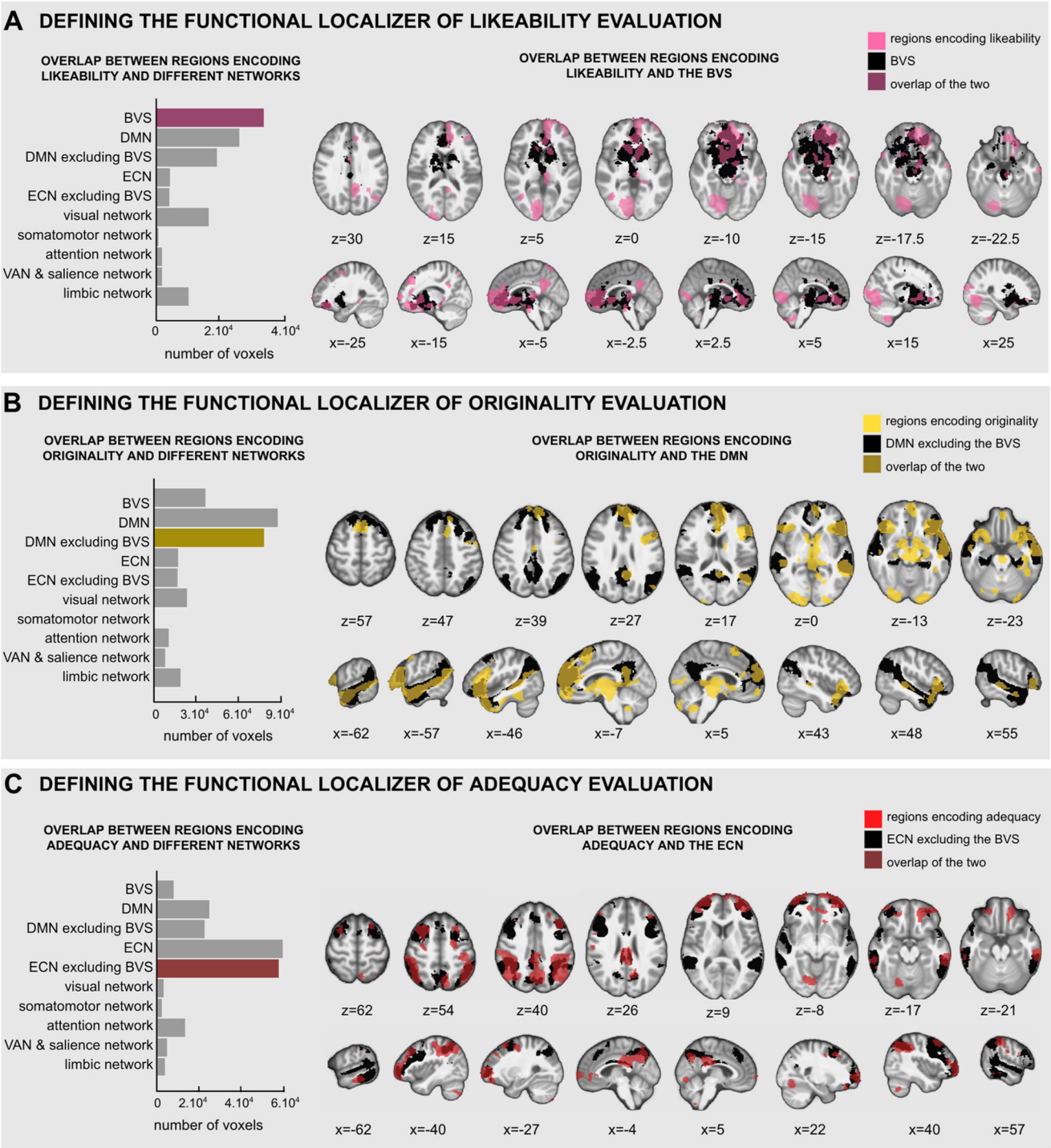


**Fig. 5 | Defining the function localizers of likeability, originality and adequacy evaluation using brain networks overlaps.** Left: Number of common voxels between the statistical maps of the parametric modulation of the likeability rating task by (**A**) likeability, (**B**) originality and (**C**) adequacy ratings, with networks of interest as defined by Yeo et al. [71]. and Neurosynth. See Table S1 for all inter-network overlaps. Note that the labels "ECN excluding BVS" and "DMN excluding BVS" are the ECN and DMN (defined by Yeo et al.[71]) after removing the regions overlapping with the BVS (defined by Neurosynth), to improve specificity. Right: Visual of the overlap between the regions encoding each dimension (color) and the most similar network (black) as defined by Yeo et a.l[71] and Neurosynth. $n = 38$.

investigated whether the originality and adequacy of ideas were represented in the brain.

### Originality and adequacy are respectively encoded in the DMN and the ECN during creative idea production

Locating brain regions encoding originality and adequacy: results from the likeability rating task: To investigate the neural representation of the originality and adequacy of ideas, we followed the same parametric modulation approach as before, this time with originality and adequacy ratings as parametric modulators of the BOLD signal in the likeability rating task: since the CES model indicates that likeability builds on originality and adequacy, we used the BOLD signal in the likeability rating task to create localizers of adequacy and originality judgments. Therefore, the GLMs we tested contained the originality (or the adequacy) of the responses as a

**Table 2 | fMRI results of the FGAT-distant**

| **parametric modulator: likeability rating** | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| mask: overlap of the likeability rating localizer with the BVS (Neurosynth metanalysis) | | | | | | | | **set *p* value = 0.007** | |
| **cluster label** | **cluster size** | **peak side** | **x** | **y** | **z** | **Z-score** | **T-value** | **peak p-value** | **cluster p-value** |
| ventral striatum | 29 | L | -14 | 8 | -14 | 04.09 | 4.64 | 0.015 | **0.0501** |
| **parametric modulator: originality rating** | | | | | | | | | |
| mask: overlap of the originality rating localizer with the DMN excluding the BVS | | | | | | | | **set *p* value = 0.0004** | |
| **cluster label** | **cluster size** | **peak side** | **x** | **y** | **z** | **Z-score** | **T-value** | **peak p-value** | **cluster p-value** |
| medial frontal pole | 20 | L | -11 | 55 | 0 | 4.1 | 4.68 | 0.044 | **0.215** |
| **parametric modulator: adequacy rating** | | | | | | | | | |
| mask: overlap of the adequacy rating localizer and the ECN excluding the BVS | | | | | | | | **set *p* value = 0.292** | |
| **cluster label** | **cluster size** | **peak side** | **x** | **y** | **z** | **Z-score** | **T-value** | **peak p-value** | **cluster p-value** |
| dorsolateral prefrontal cortex | 8 | R | 44 | 42 | 13 | 3.77 | 4.20 | 0.127 | 0.355 |
| lateral frontal pole | 8 | R | 42 | 55 | 0 | 3.30 | 3.59 | 0.445 | 0.355 |
| mask: 10-voxel radius sphere centered on the 1st peak activation in the adequacy rating localizer (x = -6 y = -70 z = 38) | | | | | | | | no suprathreshold clusters | |
| mask: 10-voxel radius sphere centered on the 2nd peak activation in the adequacy rating localizer (x = -41 y = -48 z = 38) | | | | | | | | no suprathreshold clusters | |
| mask: 10-voxel radius sphere centered on the 3rd peak activation in the adequacy rating localizer (x = -36 y = 52 z = 3) | | | | | | | | no suprathreshold clusters | |
| mask: 10-voxel radius sphere centered on the 4th peak activation in the adequacy rating localizer (x = 44 y = -38 z = 43) | | | | | | | | no suprathreshold clusters | |
| mask: 10-voxel radius sphere centered on the 5th peak activation in the adequacy rating localizer (x = 34 y = 62 z = 6) | | | | | | | | **set *p* value = 0.049** | |
| **cluster label** | **cluster size** | **peak side** | **x** | **y** | **z** | **Z-score** | **T-value** | **peak p-value** | **cluster p-value** |
| lateral frontal pole | 2 | R | 39 | 58 | 0 | 3.13 | 3.37 | **0.045** | 0.035 |

The clusters listed here result from ROI-based parametric modulation analyses. ROIs are specified under the 'mask' subheading. Coordinates x, y and z refer to the Montreal Neurological Institute space. p-values are FWE ($p < 0.05$) corrected. Only the results in bold are reported in the text.

parametric modulator of a categorical boxcar regressor between the onset of the cue (i.e., the two-word association) and the first button press to provide the likeability rating.

We found a significant correlation between the associations' originality ratings and the BOLD signal in regions classically identified as parts of the DMN during the likeability rating task (Fig. 4A): the inferior frontal gyrus ([-44, 28, -4], t(37) = 11.46, $p$FWE<$10^{-16}$), the middle temporal gyrus ([-56, -15, -12], t(37) = 9.30, $p$FWE<$10^{-16}$), the dorsomedial prefrontal cortex ([-6, 48, 16], t(37) = 9.10, $p$FWE = $3.10^{-15}$), the lateral orbitofrontal cortex ([39, 30, -10], t(37) = 7.57, $p$FWE = $3.10^{-8}$), and the pre-supplementary motor area ([-4, 20, 58], t(37) = 7.48, $p$FWE = $6.10^{-7}$).

In parallel, we found a significant correlation between the associations' adequacy ratings and the BOLD signal in regions classically identified as parts of the ECN during the likeability rating task (Fig. 4A): the precuneus ([-6, -70, 38], t(37) = 7.80, $p$FWE = $3.10^{-11}$), the left and right intraparietal lobule ([-41, -48, 38], t(37) = 7.77, $p$FWE = $1.10^{-16}$; [44, -38, 43], t(37) = 7.04, $p$FWE = $9.10^{-9}$), the posterior cingulate cortex ([-6, -25, 30], t(37) = 7.59, $p$FWE = $5.10^{-9}$), the left and right lateral frontal pole ([-36, 52, 3], t(37) = 7.36, $p$FWE = $4.10^{-6}$; [34, 62, 6], t(37) = 6.24, $p$FWE = $5.10^{-6}$), and the inferior temporal sulcus ([-61, -30, -17], t(37) = 6.50, $p$FWE = $7.10^{-5}$ - see Table 1 for all significant clusters).

This aligns with our hypothesis that originality and adequacy ratings are encoded during idea evaluation, although we had no specific hypotheses regarding the respective underlying brain networks.

Identifying the DMN and the ECN: To position the identified regions in brain networks and define ROI masks, we followed the same approach as for the likeability localizer (Fig. 5). We found that the regions reflecting originality ratings mostly overlapped with the DMN (82 262 voxels out of 248 106, i.e. 33% of overlap) and that the regions reflecting adequacy ratings mostly overlapped with the ECN 59 921 voxels out of 172 662, i.e. 35% of overlap), more so than with the other 7 networks, which confirmed our visual observations (Fig. 5 and Table S1 for all overlaps). For the following analyses, we defined the DMN and ECN ROIs (the dark yellow and dark red areas in Fig. 5B and C) as the overlapping voxels between the functional localizer (the light yellow and light red areas in Fig. 5B and C) and its corresponding atlas functional network (the black areas in Fig. 5B and C) while excluding the voxels that were in common with the BVS ROI (see Methods for more details).

Encoding of originality and adequacy in the DMN and ECN respectively during creative idea production: To investigate whether the DMN and ECN respectively represented originality and adequacy during the production of creative ideas, we followed the same approach as for likeability. We performed ROI-based parametric modulation analyses during the FGAT-distant to determine whether each network encoded its associated dimension during the production of creative ideas and specify which regions in particular. The GLMs we tested contained the originality (or the adequacy) of the responses as a parametric modulator of a categorical boxcar regressor between the onset of the cue and the button press to provide the creative response.

We found a significant correlation with originality within the DMN ROI (set $p$-value = $3.10^{-4}$; see Table 2 for details). Plus, when investigating the clusters within the ROI, we found a significant correlation between the originality of the response and the BOLD signal in the medial frontal pole ([-11, 55, 0], t(37) = 4.68, peak $p$-value(FWE) = 0.044; Fig. 4B).

For adequacy, the set $p$-value was not significant ($p = 0.29$), and we found no significant correlation between the associations' adequacy and the BOLD signal anywhere at the cluster level. We resorted to using small-volume correction with a sphere centered on the five activation peaks from the localizer. We found a significant correlation between the associations' adequacy and the BOLD signal only in the sphere located in the lateral frontal pole (sphere center coordinates: [34, 62 6], peak activation coordinates: [39, 58, 0], t(37) = 3.37, peak $p$FWE = 0.045, set $p$ value = 0.049; Fig. 4B; see Table 2 for details).

Overall, these analyses first revealed that the DMN and the ECN respectively encoded originality and adequacy values during the likeability rating task, i.e. even when we had not instructed the participants to consider these two dimensions. In other words, participants monitored the association's originality and adequacy while rating how much they liked it. Second, this encoding in the DMN and ECN was found again during the creativity task, suggesting that creative idea production also involves an evaluation of the originality and adequacy of the idea on top of its subjective value.

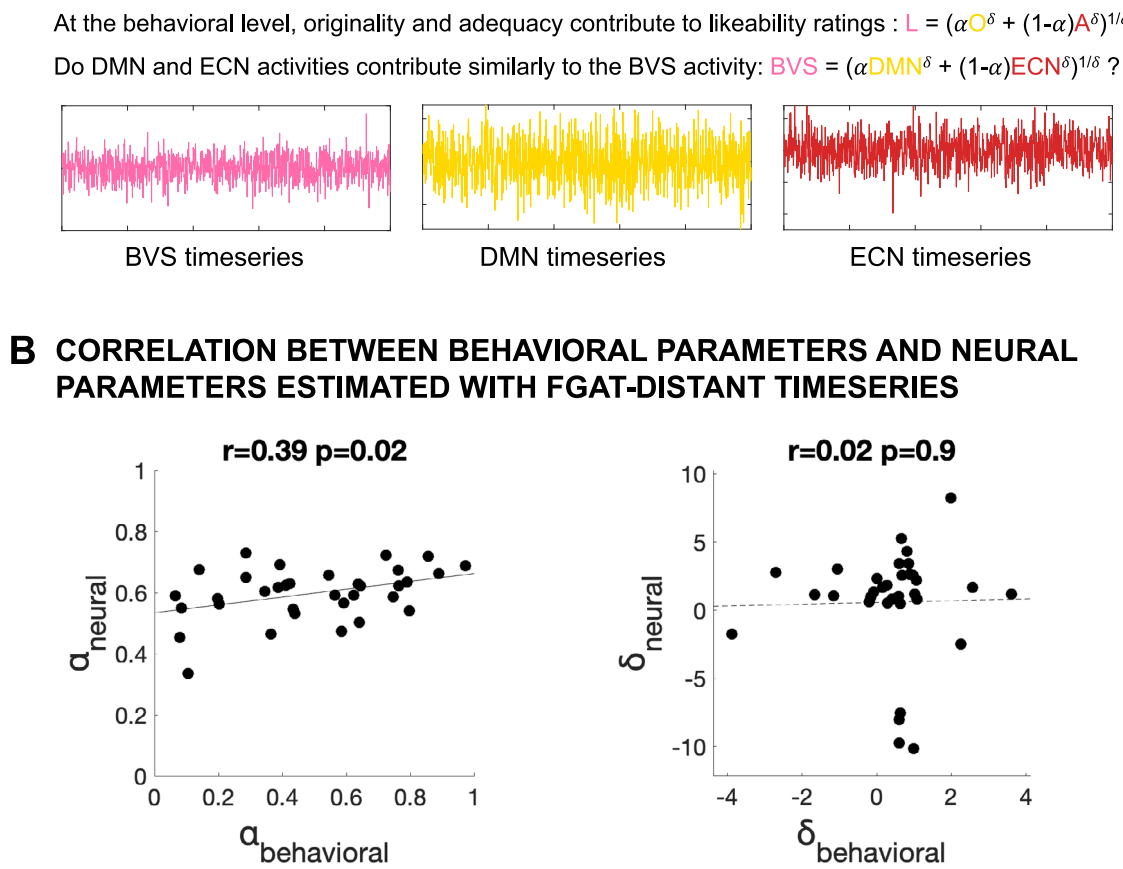


**Fig. 6 | Correlation between the parameters of the CES, estimated with behavioral and neural data. A** Schematics of the analysis approach. The timeseries depicted here for illustration purposes are simulated data. **B** Correlation between the behavioral parameters and the neural parameters estimated using the FGAT-distant timeseries. $\boldsymbol{\alpha_{behavioral}}$ and $\boldsymbol{\delta_{behavioral}}$ were estimated using the CES model fitted on the likeability, originality and adequacy ratings. $\boldsymbol{\alpha_{neural}}$ and $\boldsymbol{\delta_{neural}}$ were estimated using the CES model fitted on BVS, DMN and ECN timeseries during the FGAT-distant task. Each dot represents one participant. The solid line indicates a significant correlation ($p < 0.05$), and the dashed line indicates a non-significant correlation ($p > 0.05$). $n = 38$.

### The functional connectivity of the DMN, ECN and BVS

The results above show that originality, adequacy and likeability are respectively encoded in the DMN, ECN and BVS during the creativity task (FGAT-distant). On the other hand, we have also shown that originality and adequacy relate to likeability ratings, as captured by the CES model. Here, we wanted to test whether, during the creativity task (FGAT-distant), the contribution of ECN and DMN activities to the BVS activity followed the same relationship (i.e., the CES model) as the contribution of originality and adequacy to the likeability ratings (Fig. 6A). In other words, we wanted to check whether a connectivity pattern related to valuation was at play during the creativity task.

For each participant and for each network, we extracted the timeseries of the FGAT-distant activity. Then, we fitted the CES value function to the fMRI data, with the BVS timeseries as the dependent variable and DMN and ECN timeseries as the independent variables. This allowed us to estimate a neural $\alpha$ parameter and a neural $\delta$ parameter (Fig. 6B).

Across individuals, we found a significant correlation between the $\alpha_{\text{neural}}$ and $\alpha_{\text{behavioural}}$ parameters ($r = 0.39$, $p = 0.023$) but not between the $\delta$ parameters ($r = 0.02$, $p = 0.9$) (Fig. 6B). Together, these results suggest that the pattern of functional connectivity between the BVS, the ECN and the DMN reflects individual preferences during the creativity task.

Finally, we explored whether this result remained when looking at the activity during the other tasks and the resting-state session and found varying results (Supplementary Results and Figure S3), which we discuss in the Supplementary Discussion.

## Discussion

### Summary of findings

In this study, we used a creative word association task and several rating tasks to unveil the neurocognitive bases of the evaluation step of creativity. Our behavioral results confirm our prior findings that, during creativity, the subjective valuation of ideas (i) depends on the originality and adequacy of the ideas, (ii) energizes the production of ideas and (iii) varies across individuals and relates to interindividual differences in creative abilities. This is supported by our neural findings showing that, during creativity, (i) subjective values are encoded in the BVS, (ii) originality and adequacy are encoded in the DMN and the ECN respectively, and (iii) the DMN and ECN contribute to the BVS activity in a fashion that mirrors behavioral preferences and supports interindividual variability. Overall, our results reveal the underlying neurocognitive mechanisms of the evaluation step of creativity and validate our hypothesis that the creative process is guided by the subjective valuation of ideas and the key dimensions of a creative product.

### The role of the BVS in valuation and decision-making

We showed that the subjective value of an idea is represented in individuals' BVS when we explicitly asked them to rate how much they liked an idea, but also when they focused on producing a creative idea. This latter finding helps with the interpretation of a previous behavioral result, regarding the relationship between likeability ratings and response times during the FGAT-distant. Since the likeability ratings correlate with the BVS activity during the FGAT-distant (i.e., before the rating task), it supports the hypothesis that likeability influences the production speed of creative

answers, rather than the alternative - that is, individuals preferring ideas because they found them faster.

Besides that, our neuroimaging findings are consistent with two BVS properties reported in the decision-making literature[49]. First, the involvement of the BVS in the valuation of creative ideas is consistent with the generic property of the network: some have demonstrated that the BVS could reflect the subjective value of different kinds of items, no matter their nature[51,72]. This property aligns with the neural common currency framework, which states that the values of different and disparate types of items can be represented in the same brain regions and allows for their comparison[52]. Second, the valuation of ideas in the BVS during creative production is consistent with the automatic property of the network: the BVS can reflect an item's value even when an individual is not explicitly instructed to rate how much they like it. For instance, when an individual is focused on another task such as passively viewing a stimulus[53], determining a person's age[49,51] - or in our case, producing a creative word association - the BVS still reflects how much individuals like the item they are processing.

In parallel, the role of the BVS - and, in particular, the ventromedial prefrontal cortex (vmPFC) - has been challenged by recent studies. They argue that rather than representing one's subjective value for the object they are processing[48,49,51,73,74], the BVS could be representing slightly different variables, such as one's goal value[75–81] or one's confidence in the rating[82–84]. In the current study, control analyses found no significant evidence indicating a confound with confidence. On the other hand, our study design does not allow us to differentiate between subjective and goal values. Subjective value differs from goal value in that the former is generally stable, regardless of the context, while the latter highly depends on one's current goal. For example, the subjective value of a boat might always be minimal for someone not interested in traveling on the sea, but its goal value would increase dramatically if that person were stranded on a desert island.

This distinction might be relevant to the current study, where we observe that likeability ratings are represented in the BVS during the "distant" condition of the FGAT (where we instructed participants to give a creative association) but not during the "first" condition (where we instructed participants to give the first association that came to their mind). During the likeability rating task, the instruction was to rate the associations in the context of the FGAT-distant condition (where the goal was to be creative). As a result, the ratings given by the participants may reflect goal values more than subjective values. If so, since the goal to think creatively is similar in the FGAT-distant condition and the likeability rating task, the goal value would remain the same, which might explain why we observed its encoding in the BVS during the FGAT-distant. In contrast, the goal in the FGAT-first condition does not involve creativity and thus differs from that of the likeability rating task. This could explain our null result regarding the correlation between the brain activity during the FGAT-first and the likeability ratings, as these ratings might actually represent the goal value for a goal that has yet to be established.

Some studies also linked the BVS to the encoding of yet other dimensions. They observed that novelty processing was related to increased activity within the ventral striatum[85], the substantia nigra and the ventral tegmental area[85–88]. This heightened activity was viewed as a signal for learning when it was associated with enhanced synaptic plasticity in the hippocampus[89]. However, these studies did not assess the relationship between BVS and novelty processing with creative functions.

### The role of the BVS in creativity

Despite the widely acknowledged impact of motivation on creativity, the role of the BVS has been largely overlooked, with only a few studies debating its involvement in creativity, mainly for originality or adequacy processing. Huang et al.[57] showed that, during creative judgment tasks, BVS regions, such as the caudate nucleus and the substantia nigra, represented originality. In a later study, however, they showed that other regions from the BVS, namely the ventral striatum and orbitofrontal cortex, actually represented adequacy[90]. Similarly, Matheson et al.[91] found that medial prefrontal cortex (mPFC) regions, which belong to both the BVS and the DMN, supported adequacy processing. Additionally, other studies found that functional connectivity in subcortical regions of the BVS, such as the putamen[92], and grey matter volumes in the striatum and ventral tegmental area[56,58] were all related to better divergent thinking abilities. The involvement of the BVS in creativity is also corroborated by investigations at the molecular level, such as Aberg et al.[59] who showed that individuals with decreased dopamine in the right hemisphere scored better in creativity tasks requiring remote associations (Alternative Uses Task and Remote Associates Test).

An overarching complication when studying motivation and creativity is the anatomical similarity between the BVS and the DMN. In the current study, we assessed the overlap of the DMN (as defined by Yeo et al.'s seven networks atlas[71]) and the BVS (as defined by the "reward" term-based meta-analysis of the Neurosynth platform, https://neurosynth.org/[92]) and quantified a total of 15 671 voxels, a volume which amounts to 6% of the DMN, or 14% BVS (Table S1). This anatomical similarity can be a common issue in the literature where some creativity studies label "DMN" regions that would be labeled "BVS" by neuroeconomists. It will be critical for future research to be aware of this overlap and eliminate this source of confusion, for instance, by using localizers, as we did in the current study.

Regardless, while the literature does not clearly establish how the BVS and its related dopaminergic pathways are involved in creativity, it converges towards confirming their central role[59,93]. Nevertheless, the present study demonstrates the involvement of the BVS in encoding an idea's subjective value during the creative process. Previous findings linking the BVS with originality or adequacy processing do not offer mechanistic explanations and may be consequences of the correlation we observed between likeability and the two creative dimensions. In contrast, our innovative design helps break down the evaluation step of creativity into the monitoring of ideas' originality and adequacy and the valuation of ideas, clarifying the specific role of the BVS in this process. We also challenge the established roles of the DMN and ECN by investigating the encoding of originality and adequacy, whose neural correlates were unclear.

### The roles of the DMN and ECN in creativity: revisiting the classical generative and evaluative distinction

The current literature broadly establishes that the DMN is involved in the generation phase of creativity and helps form new associations in a spontaneous manner. In parallel, the ECN is reported to support the evaluation step of creativity in a goal-directed and controlled manner. This one-to-one matching of the dual-process of creativity to a dual-network organization is supported by many studies suggesting that the DMN is involved in idea generation[21,91,94–97]; that the ECN is involved in idea evaluation[21,90,91,94–96,98]; and that creativity relies on these networks' cooperation[28,40,45,47,99,100].

However, some findings challenge the exact mapping of the dual-process of creativity to its dual neural basis. First, several studies, like ours, report that idea evaluation can be partly supported by the DMN[39,94,95]. Some, even more in alignment with our results, precise that the DMN is related to originality processing[96,98]. If we consider that achieving originality requires making new associations, it makes sense that the DMN, which has consistently been associated with associative processes during generation, would encode this dimension during evaluation. However, other studies slightly diverge from our own and find that the DMN supports the processing of adequacy[57]. Regardless of some diverging results that could be due to tasks and analysis differences, many studies challenge the now common conception that the DMN is only involved in idea generation. Second, other studies, like ours, find that the involvement of the ECN is indeed related to the evaluation of an idea, particularly its adequacy dimension[41]. If we consider that achieving adequacy requires exploring candidate ideas and sorting through them to find the most adequate one, it makes sense for this to be associated with cognitive control processes and, therefore, supported by the ECN. However, contrary to our results, others find that the ECN supports the evaluation of an idea's originality[90,91,96].

In summary, although the studies cited above lack perfect convergence, they question the neural bases of the dual process model of creativity and

suggest that (as in our study) evaluation processes rely on both the DMN and ECN. Our study advances this understanding by providing precise, mechanistic evidence of the DMN and the ECN encoding originality and adequacy, offering neural support for the intuitive but largely unproven idea that creativity involves evaluating these two dimensions.

### The complementary roles of the BVS, the DMN and the ECN in creativity

More and more studies question the computations at play during the evaluation phase of the creative process, highlighting that this step is often simplified and overlooks the importance of emotions, motivation or valuation[28,35,90].

Based on Huang et al.[90] results, which showed that originality and adequacy processing relied respectively on regions of the ECN and the BVS, Kleinmintz et al.[28] attempted to dissociate the evaluation process into "monitoring" and "valuation" subprocesses. They propose that the assessment of originality requires cognitive control akin to monitoring and that, in contrast, the assessment of adequacy is linked to emotional and motivational processes akin to valuation.

Our view partly differs from theirs. First, we dissociate evaluation into monitoring and valuation as well, but align monitoring with the evaluation of adequacy and originality (i.e., the monitoring of the dimensions necessary for a creative goal) and valuation with the assignment of a subjective value that combines the monitored adequacy and originality. Second, we show that the ECN encodes adequacy, not originality and that the DMN encodes originality. Finally and most importantly, we show that the BVS underlies the valuation of ideas, not the processing of adequacy. These diverging results may stem from variations in task design: in the study conducted by Huang et al.[90] participants reviewed creative solutions to riddles and assessed whether they were adequate or not, while here, participants reviewed creative word associations and assessed how much they liked them, and how adequate and original they were. Additionally, methodological differences might contribute to divergences in results: Huang et al.[90] used high versus low adequacy and originality contrasts, while we used parametric modulation based on continuous adequacy and originality ratings.

Overall, we agree with Kleinmintz et al.[28] that what is commonly labeled evaluation should be subdivided into monitoring and valuation subprocesses. We propose an alternative model in which originality and adequacy are monitored and integrated during valuation, which guides the final selection step during idea production.

A remaining important question is whether evaluation's monitoring and valuation subcomponents are consistently both at play during creativity. In the current study, we show that both valuation and monitoring of originality and adequacy are at play during the production of creative word associations. This was generalized to the production of alternative uses for common objects and the production of creative drawings from incomplete shapes in another study from the lab[101]: in these tasks too, participants' likeability ratings built on the creative products' originality and adequacy, and their production speed was predicted by how much they liked them. This indicates that valuation might generally be involved in the creative process, regardless of its domain. To generalize these results further, future creativity studies should include likeability rating tasks in their design to assess the participants' subjective values and check if they correlate with behavioral and neural variables (such as the response times and the BVS activity used in the current study).

Additionally, the current study finds that subjective value scales with originality and adequacy ratings. First, this result means that valuation builds on originality and adequacy, the two key dimensions in the definition of creativity. Second, it adds to prior research on multi-attribute value-based decision-making. This literature studies choices that depend on the value of several criteria: for example, food choices can depend on taste and health attributes. However, the brain networks implicated in multi-attribute value-based decision-making are still unclear. The neural representation of attributes may be contingent upon their nature, and in some cases, attributes may not be represented at all in the brain. For instance, in the context of food choices, regions such as the dorsolateral prefrontal cortex (dlPFC) are associated with the healthy attributes of food, while the ventromedial prefrontal cortex (vmPFC) is linked to the tasty attributes[102,103]. Similarly, in the realm of social decision-making, brain regions like the temporoparietal junction (TPJ) or the medial prefrontal cortex (mPFC) are implicated in processing social attributes, while the ventral striatum plays a role in self-related attributes[104–107]. Yet, the neural representation remains uncertain for risky choices and perceptual attributes like shape or colors, with some evidence for attribute representation in the dlPFC and combined value representation in the vmPFC[108]. Notably, several studies do not report specific activity directly linked to attributes. Instead, some suggest that attributes are represented in the same areas as the combined value[109]. Certain studies, such as the one by Hunt et al.[110] have focused on comparing attributes in the intraparietal lobule, while others propose the involvement of the dlPFC in attribute representation and comparison[111]. It is essential to highlight that these studies often report brain activities in specific regions rather than brain networks, making direct comparisons challenging. For example, the TPJ and mPFC, both part of the DMN, substantially overlap with the social network[112].

Here, we find that during creative valuation, the attributes of creative ideas (originality and adequacy) are separately represented in networks consistently reported in creativity research (in the DMN and ECN, respectively). These findings illustrate that attributes can be represented in different brain regions and that these regions (here, creativity networks) could be relevant to the specific nature of the attributes under consideration (here, creative dimensions).

Finally, the significant correlation between the neural CES parameter associated with DMN and ECN activities ($\alpha_{neural}$) and the behavioral CES parameter related to originality and adequacy assessments ($\alpha_{behavioral}$) suggests a neural basis for individual preferences in creativity. This finding underscores the dynamic interplay between cognitive networks involved in creativity and subjective valuation, offering valuable insights into the neural underpinnings of preferences and decision-making in creativity.

### The link between idea valuation and creative abilities

The present study not only shows that individuals combine the dimensions of originality and adequacy into a subjective value, but also demonstrates that the way they combine them (as estimated by the α and δ parameters of the CES model) relates to their creative abilities: giving more weight to an idea's originality over its adequacy (i.e., having a high α) and preferring a trade-off of these two dimensions rather than an extreme of either (i.e., having a low δ) correlated with higher scores in creativity tests.

This finding replicates Lopez-Persem et al.[36] and suggests, together with other studies[7,24,25,27], that the evaluation step is essential in the creative process. For instance, previous results showed that an overly stringent evaluation inhibited creativity[5,96]. It was also illustrated by Kleinmintz et al.'s study[26], in which they found that improviser musicians were more lenient than non-improviser musicians when rating creative ideas and that this helped them achieve greater creativity.

In parallel, Diedrich et al.[31] showed that scores in a creativity task were correlated to the idea's originality and not to its adequacy. Adequacy did predict creativity but only in very original ideas, as the interaction of originality and adequacy explained additional variance of the creativity scores. Here, we computationally quantify similar effects with the α and δ parameters: higher creativity is achieved by giving greater importance to originality (high α) but still preferring a trade-off of originality and adequacy over extremely original but poorly adequate options (low δ).

### Limitations

The present study identifies that likeability, originality and adequacy are respectively encoded in the BVS, DMN and ECN as part of an evaluation process, both during a rating task and a creative production task. However, these interpretations are based on parametric modulation analyses, i.e.

correlations, which strictly indicate brain regions where the BOLD signal increases as a function of a given factor (the parametric modulator) that varies across trials. Thus, instead of concluding that "the likeability of ideas is encoded in the BVS during the rating task", a more conservative interpretation would be "the greater the likeability rating, the greater the BOLD signal in regions of the BVS". We cannot exclude that another variable might mediate these correlations.

In other studies, higher activity in the BVS during creativity judgments was interpreted as emotional arousal[57] or motivation[94]. Additionally, Wu et al.[98] proposed that higher activity in the DMN during creativity judgments may stem from participants generating alternatives while viewing the one in front of them. Our analyses cannot rule out these alternative explanations. Yet, if one could argue that, indeed, the DMN might be more active in the FGAT-distant due to alternative ideas generation, the fact that we also observed its activity in the likeability rating task, where no generation occurs, suggests that the DMN supports idea evaluation.

## Conclusion

In the current study, we clarify the cognitive and neural mechanisms involved in idea evaluation, a critical but long-overlooked step of the creative process. We combined creativity and neuroeconomics methods to show that valuation participates in idea evaluation, meaning that individual preferences influence the creative process. These preferences depend on a trade-off of originality and adequacy, which are primary creativity criteria. At the neural level, we demonstrate the involvement of the DMN and ECN in the evaluation of originality and adequacy, respectively, and add the BVS to the incomplete neural framework of creativity.

### Perspectives

Future research should delve deeper into how the DMN, ECN and BVS interact during idea evaluation. In parallel, to further understand the decision-making processes of creativity, it would be helpful to investigate the selection step of creativity, i.e., how individuals choose the best candidate idea. Like monetary or food choices, creative choices are likely driven by subjective values, as we suggest in Lopez-Persem et al.[36] In our model, the candidate ideas are compared, and the one with the highest value has a higher probability of being selected. An additional hypothesis of our general framework, to be tested in future studies, is that if none of the candidate ideas are satisfactory, the process starts again with new idea generation. Finally, the model we propose calls for the study of the neural bases of the selection process. It also paves the way for investigating suboptimal or biased choices, which are a central issue considering the growing literature on cognitive biases in creativity research[5,113–117].

## Methods

### Participants

An official ethics committee approved the study (CPP Ouest II – Angers). We recruited and tested forty participants at the CENIR platform of the Paris Brain Institute (ICM). All participants were French native speakers, right-handed, with correct or corrected vision, and no neurological or psychological disease history. We excluded two participants from analyses, one because we interrupted MRI scanning after they suffered from a claustrophobic episode and one because of a technical issue during MRI scanning. The final sample comprised thirty-eight participants (19 women, age = 26.5 ± 0.7 (M ± SEM), level of education = 16.7 ± 0.4 years (M ± SEM)). They were paid 110€ for a four-hour testing session that included MRI scanning and cognitive and creativity tests inside and outside the scanner. All ethical regulations relevant to human research participants were followed.

### Experimental design

After giving informed consent, participants completed an MRI session composed of an anatomical T1 scan, four task-based fMRI scans (while performing an FGAT-first, FGAT-distant, a likeability rating task and a choice task) and a resting-state fMRI scan. Participants then completed an originality and adequacy rating task and a battery of creativity tests outside the MRI. We used Matlab (MATLAB. (2020). 9.9.0.1495850 (R2020b). Natick, 624 Massachusetts: The MathWorks Inc.) to program the tasks and the Qualtrics software (Qualtrics, Provo, UT, USA. https://www.qualtrics.com) to implement creativity tests.

**Free Generation of Associates Task**. The Free Generation of Associates Task (Fig. 1B) is a remote thinking task reported to capture critical aspects of creativity[63,64]. In our experimental design, the task comprised two successive conditions, each comprising 5 training trials and 62 randomized test trials. Each trial displayed a fixation cross for 1.6 to 3.2 s, jittered with a uniform distribution. Then, a cue word was displayed. In the "first" condition, participants had up to 10 s to give the first word that came to mind after reading the cue word (for example, participants might answer "father" to the cue word "mother"). In the "distant" condition, they had up to 20 s to provide a distant word, so the cue and response words would result in a creative word association (for example, participants might answer "nature" to the cue word "mother"). Participants pressed the response button with their right index finger when they had an answer in mind. Then, they said the response word out loud into a microphone. The experimenter typed the response and displayed it on the participant's screen. Participants then had the opportunity to repeat their response in case the experimenter had misheard (by pressing the response button again with their right index finger) or to validate the response (by pressing the validation button with their right ring finger). The exact instructions given to the participants are in the Supplementary Methods.

**Rating task**. We split the rating task in two: a likeability rating task performed inside the scanner and an originality and adequacy rating task performed outside the scanner. Each rating task counted five training trials and a variable number of test trials (M ± SEM = 145 ± 2), depending on the validity of the participant's FGAT answers (see Supplementary Methods for more details). For both rating tasks, the trials were composed of the following: 66% of the trials were evenly made up of the participant's valid FGAT-first and FGAT-distant associations; 27% of the trials were a sample of frequent and rare responses taken from another study using the same FGAT cue words; and finally 7% of the trials were responses that were totally unrelated to the cue words they were associated with (for example: 'cow' and 'inverse'). We added these associations to cover a wide range of adequacy and originality ratings, ensuring a robust estimation of likeability with sufficient statistical power.

The first rating task, i.e. the likeability rating task (Fig. 1C), was performed in the MRI scanner. Participants had to indicate how much they liked the association or how satisfying it was in the context of the FGAT-distant condition. In this task, each trial started with a fixation cross displayed for 1.6 to 3.2 s (to meet fMRI constraints, display times varied between trials and were determined by a random jitter with a uniform distribution). Then, a cue word (for example, "mother") was displayed alone for 0 to 0.5 s. Then, a response appeared under the cue word. This association (for example, "mother-nature") remained alone for 1.6 to 3.2 s. Then, a rating scale appeared below the association. The rating scale's low to high values were represented from left to right, with the words "not at all" and "very much" located at the extremities but without any numerical values. A segment indicated the middle of the scale, which was divided into 101 hidden steps, later converted into ratings ranging between 0 and 100. We displayed a heart icon (for likeability) under the rating scale. To provide a rating, participants had to move a cursor by pressing the first and second MRI response buttons with the right index and right middle finger and then validate the rating by pressing the third button with their right ring finger. Once the response was validated, the subsequent trial began. The exact instructions given to the participants are in the Supplementary Methods.

The second rating task, i.e. the originality and adequacy rating task (Fig. 1D), was completed after the choice task and outside the MRI. Again, each trial displayed a two-word association (for example, "mother-nature")

and asked participants to subsequently rate how original and how adequate it was. The order of originality and adequacy ratings was counterbalanced across trials. Each trial started with the screen displaying a cue word. After 0.3 s, a response word appeared under the cue, and after 0.4 s, a scale appeared under the two-word association. The scale was similar to the likeability rating scale except for the icon displayed underneath: a star for originality and a target and arrow for adequacy. Participants rated the association by moving the slider across the scale using the left and right arrows of the keyboard and validated by pressing the spacebar. Then, a new scale appeared underneath the same association, indicating the second dimension (adequacy or originality). Participants provided their ratings similarly and then started the subsequent trial.

We told participants to use the whole scale throughout the task. No time limit was applied, but the instruction was to respond spontaneously. The exact instructions given to the participants are in the Supplementary Methods.

**Choice task**. The choice task was performed in the MRI and comprised five training trials and an average of 187 randomized test trials (the exact number varied between participants depending on the variability of their FGAT answers and ratings). Each trial displayed two possible responses to an FGAT cue, with the cue word on the top of the screen and the two response options words below, on the same horizontal line (for example, "mother" was displayed on top of "nature" and "father"). Participants had to choose which of the two responses would have resulted in the best association with the top cue in the FGAT-distant condition. No time limit was applied, but the instruction was to respond spontaneously. Note that the results of this task go beyond the scope of this paper and will instead be the focus of a future publication. The exact instructions given to the participants are in the Supplementary Methods.

**Battery of creativity tests**. The battery comprised a daily-life creativity questionnaire, self-reports, and established creativity tasks belonging to three main frameworks of creativity—the divergent thinking approach, the associative theory, and the insight problem-solving approach.

Inventory of Creative Activities and Achievements: the inventory of creative activities and achievements was first designed by Diedrich et al.[118] as an ecological measure of creativity. It focuses on eight domains of creativity (literature, music, crafts, cooking, sports, visual arts, performing arts, and science and engineering). For each domain, participants reported their level and their frequency of engagement over the past ten years. They also specified their level of creative accomplishment in each of these domains. We computed the ICAA score as the mean of the creative activities score and the creative achievements score. First, for the creative activities score, we summed the frequency at which the participant had engaged in each of the eight creativity domains using an ordinal scale ranging from 0 (never) to 4 (more than ten times). Then, for the creative achievements score, we summed the participants' levels in each of the eight creativity domains. These ranged from 0 ("never engaged in this domain") to 10 ("I have already sold some of my work in this domain"). Finally, we averaged the two scores into a unique ICAA score per participant.

Drawing task: the drawing task[119] comprised a training trial and 12 test trials. We instructed participants to include incomplete shapes in creative drawings. The 12 incomplete shapes were relatively similar: they were all composed of 4 lines, either straight, curved or right-angled, and exhibited horizontal symmetry (cf.[119] for an example). Participants completed the 12 drawings at their own pace. Then, they reviewed each drawing and gave each one a verbal title. We followed the consensual assessment technique that is advised for scoring such a creativity task[33]. We designated four judges from the lab who were familiar with creativity judgments but had not seen the data before the rating. For each of the twelve shapes, the judges first saw the incomplete shape, then each participant's drawing using that shape, in a blind and randomized order. Then, the judges gave each drawing a rating between 0 (not creative at all) and 4 (extremely creative). We tested the inter-judge independent rating consistency using an intraclass correlation analysis, which gives the mean correlation between the judges' ratings ($r = 0.8239 \pm 0.02$; $M \pm SEM$).

Combination of associates task: Bendetowicz et al.[120] first developed the combination of associates task (CAT) by adapting the remote associates test from Mednick[66]. After five training trials, participants performed 40 randomized test trials, each displaying three cue words with no apparent association: they had to find a fourth word that linked all of them (for example, the link between "map", "dig", and "discover" is "treasure"). The CAT score used for the analyses was the participants' total number of accurate answers.

Associative fluency task: in this task, adapted from Benedek et al.[121], participants had two minutes to type as many words related to a cue word as possible. We chose this associative design over simple category or letter fluency tasks to measure creative abilities on top of fluency. We selected six cue words ("garden", "wine", "rock", "opinion", "call", and "finger") among the FGAT cues for their diversity in steepness. The steepness is the ratio of the frequency of the most frequent answer over the frequency of the second most frequent answer: a steep cue word (e.g., "cat") has a very common first associate ("dog"); a flat cue word has several equally-common first associates. We measured typing speed and accounted for it in the fluency analysis. The score used for the analyses was the total number of answers given to a cue word, averaged across all cue words.

Participants also performed an alternative uses task (AUT), a self-report of their creativity level (i.e. "How creative are you on a scale of 1 to 100?") and a questionnaire regarding their preferences in creativity (e.g., "Would you prefer an idea to be useful or novel?") that were not analyzed or used in this paper but will be part of future papers.

### Behavioral analyses

We performed all the analyses using Matlab (MATLAB. (2022). 9.12.0.2009381 (R2022a). Natick, Massachusetts: The MathWorks Inc.).

**Differences in originality, adequacy and frequency between FGAT-first and FGAT-distant responses**. For each individual, we computed the mean of originality and adequacy ratings separately for items of the FGAT-first and FGAT-distant conditions. Then, we compared the difference in adequacy between the two tasks to their difference in originality using a paired two-tailed t-test at the group level. We computed the frequency of association of all FGAT responses using the French database Dictaverf (http://dictaverf.nsu.ru/)[65]. This database is built on spontaneous associations provided by at least 400 individuals in response to 1081 words (each person saw 100 random words). Frequencies were log-transformed to take into account their skewed distribution toward 0. We compared the frequency of the FGAT-first responses and the FGAT-distant responses at the group level using a paired two-tailed t-test.

**Relationship between FGAT response times and likeability ratings**. We defined participants' response times in the FGAT as the time between the cue display and the participants' first button press, which indicated they had an answer in mind. Then, at the individual level, we performed a linear regression of response times against likeability ratings, standardized between participants. We did two distinct regressions for the FGAT-first and FGAT-distant conditions. At the group level, we performed a one-sample, two-tailed t-test on the regression coefficient to test the significance of the effect of likeability on response times. We also tested the difference in the effect of likeability on response times between the two FGAT conditions using a paired two-tailed t-test on the regression coefficients. Note that all variables were standardized at the subject level, and non-standardized data are depicted in figures.

We then performed a control analysis to check that response times were still predicted by likeability ratings even after controlling for a confound with confidence (reported in the Supplementary Results). Note that, as we did not collect confidence ratings, we used squared likeability ratings

as a proxy for confidence[67,68] (see Supplementary Methods for more details on this analysis).

**Relationship between likeability, originality and adequacy**. We fitted the participants' rating task data at the individual level with the Constant Elasticity of Substitution utility function[69,70], which provided the best fit in our previous study using the same experimental design[36].

$$Likeability\ rating_i = (\alpha \times Originality\ rating_i^{\delta} + (1-\alpha) Adequacy\ rating_i^{\delta})^{\frac{1}{\delta}}$$

"i" stands for one association, α stands for the weight given to originality ratings, and δ stands for the preference for extremes in originality or adequacy as opposed to a trade-off of those dimensions.

We used the Matlab VBA toolbox (https://mbb-team.github.io/VBA-toolbox/), which implements Variational Bayesian analysis under the Laplace approximation[122,123] to fit the model to individual data and estimate individual parameters. We tested for significance the estimated parameters of the CES model at the group level using one-sample two-tailed t-tests against their prior values (0.5 for α and 1 for δ).

**Creative abilities analyses and canonical correlation**. We pooled the scores for the associative fluency task, the Combination of Associates Task (CAT), the drawing task and the Inventory of Creative Activities and Achievements (ICAA) (see Methods for more information) into a "creativity scores" set and the α and δ parameters into a "valuation parameters" set. Then, we performed a canonical correlation analysis of these two sets. Canonical correlation analysis identifies the shared variance between two datasets, summarizing it as canonical variables. They are organized based on the strength of correlations between the two datasets. Here, we examined the correlation between the canonical variables in each dataset and presented their loadings, i.e. the contributions of each creativity score and each evaluation parameter to their respective canonical variable. Correlation coefficients were tested for significance using Bartlett's modified chi-squared tests of the Matlab "canoncorr" function, and the variables' contributions were tested for significance using the two-tailed t-test of the Matlab "corr" function.

### MRI data acquisition and preprocessing

**Scanning parameters**. We acquired neuroimaging data on a 3 T MRI scanner (Siemens 3 T Magnetom Prisma Fit) with a 64-channel head coil.

Four functional runs were acquired (FGAT-first, FGAT-distant, likeability rating task, choice task) using multi-echo echo-planar imaging (EPI) sequences. The number of whole-brain volumes per run varied between tasks but also between participants since all tasks were self-paced (mean, [min;max]: FGAT-first = 379, [317;454], FGAT-distant = 612, [359;856]; likeability rating task = 800, [650;1012] and choice task = 662, [525;821]). We did not record any dummy scans and, therefore, did not discard any volumes. The functional runs used the following parameters: repetition time (TR) = 1660 ms; echo times (TE) for echo 1 = 14.2 ms, echo 2 = 35.39 ms, and echo 3 = 56.58 ms; flip angle = 74°; 60 slices, slice thickness = 2.50 mm; isotropic voxel size of 2.5 mm; Ipat acceleration factor = 2; multiband = 3; and interleaved slice ordering.

After the EPI acquisitions, we acquired a T1-weighted structural image with the following parameters: TR = 2300 ms, TE = 2.76 ms, flip angle = 9°, 192 sagittal slices with a 1-mm thickness, isotropic voxel size of 1 mm, Ipat acceleration factor = 2.

Finally, we added a resting-state fMRI session of 10 min (360 volumes) with the same acquisition parameters as the task runs.

**Preprocessing**. We performed the preprocessing of the on-task fMRI data separately for each run and the resting-state data using the afni_proc.py pipeline from the Analysis of Functional Neuroimages software (AFNI; https://afni.nimh.nih.gov). The different preprocessing steps of the data included slice timing correction and realignment to the first volume (computed on the first echo). We then combined the preprocessed data using the TE-dependent analysis of multi-echo fMRI data (TEDANA; https://tedana.readthedocs.io/), version 0.0.9a1[124,125]. We co-registered the resulting data on the T1-weighted structural image using the Statistical Parametric Mapping (SPM) 12 package running in MATLAB (MATLAB R2017b, The MathWorks Inc., USA). We normalized the data to the Montreal Neurological Institute template brain using the transformation matrix computed from the normalization of the T1-weighted structural image with the default settings of the computational anatomy toolbox (CAT 12; http://dbm.neuro.uni-jena.de/cat/)[126] implemented in SPM 12.

**fMRI analyses**. Parametric modulation analyses: We entered the resulting normalized data from the task-based fMRI in generalized linear models (GLMs) in SPM. In this analysis, we entered 24 motion parameters and 42 physiological noise parameters as confounds regressed from the BOLD signal. In this preprocessing analysis, we used the Matlab PhysIO Toolbox[127] (version 5.1.2, open-source code available as part of the TAPAS software collection: https://www.translationalneuromodeling.org/tapas[128]) to generate nuisance regressors for the GLM. The regressors were composed of (i) physiological (cardiac pulse and respiration) recordings used to generate RETROICOR[129] regressors, (ii) White Matter (WM) and CerebroSpinal Fluid (CSF) masks from the anatomical segmentation used to extract components from compartments of non-interest using Principal Component Analysis (PCA) and (iii) motion parameters, composed of standard motion parameters, first temporal derivatives, standard motion parameters squared, and first temporal derivatives squared. Outlier volumes were eliminated when the Framewise Displacement (FD) exceeded 0.5 mm. Then, we used GLMs to explain preprocessed time-series at the individual level.

Modeling the neural encoding of likeability during the likeability rating task (functional localizer): We applied the first, second and third models (GLM1, GLM2 and GLM3) to the likeability rating task. They all included a boxcar function capturing the signal between the cue display and the first button press to move the rating scale slider (one event per trial). GLM1's parametric modulator was the likeability rating of the response, while GLM2 and GLM3 were parametrically modulated by the originality and adequacy ratings, respectively. Note that GLM2 and GLM3 parametric modulators (originality and adequacy ratings, respectively) were not orthogonalized for the other dimension (meaning adequacy and originality, respectively), but control analyses with orthogonalization led to similar results.

Modeling the neural encoding of likeability during the FGAT-distant: The fourth, fifth and sixth models (GLM4, GLM5 and GLM6) were similar to the first three models, except that we applied them to the FGAT-distant. They all included a boxcar function capturing the signal between the cue display and the first button press, signaling that the participant had found a response to the cue (one event per trial). GLM4's parametric modular was the likeability rating of the response, while GLM5's and GLM6's were the originality and adequacy ratings, respectively. Note that not all 62 FGAT-distant trials were included in this analysis: depending on the participant, the number of included trials ranged between 34 and 59, as some associations had not been rated after controlling for FGAT-first and FGAT-distant similarity (see Supplementary Methods for more details).

We convolved the regressors for all models with a canonical hemodynamic response function. The resulting masks were identified in statistical parametric maps (SPMs) at a threshold of $p < 0.05$, controlling for family-wise errors (FWE) at the cluster level. For GLMs 1 to 3, we found significant clusters at the whole brain level. For GLMs 4 and 5, we found significant clusters using small volume correction (SVC) with the overlap of the localizer and the networks of interest. For GLM6, SVC were spheres centered on the peak activation sites from the results of GLM3.

Functional localizers' similarity with networks of interest: To compare our resulting maps to our networks of interest, namely the BVS, the DMN and the ECN, we quantified the number of overlapping voxels between the SPMs of GLMs 1 to 3 and each of Yeo et al.[71] 7 intrinsic functional networks in the

MNI152 referential. Since Yeo's atlas does not include the BVS (1) we used the term-based meta-analysis of the Neurosynth platform (https://neurosynth.org/)[92] and "atlas BVS network" is the result of an automated meta-analysis of all studies in the Neurosynth database whose abstracts include the term "reward" at least once and (2) to improve specificity, we also computed "DMN excluding BVS" and "ECN excluding BVS" networks, by taking the ECN and DMN of the Yeo atlas and removing the regions overlapping with the BVS (defined using Neurosynth). Before quantifying the overlaps, we verified that the atlas masks aligned with our results using the SPM checkReg function and applied it to our data with the ImCalc tool in SPM12.

Fitting the CES to the FGAT-distant timeseries: To investigate the interaction between the BVS, ECN and DMN during the FGAT-distant, we extracted the timeseries following a procedure previously used in the lab[19]. To covary out the task-related signal from the FGAT-distant run, we entered the preprocessed fMRI data in a GLM in SPM. We regressed out the onsets of each task-related event (cross onset, cue onset, first press event, validation event) from the BOLD signal. We then standardized and detrended the residuals of the GLM and concatenated them to obtain timeseries for each voxel. We averaged the timeseries across all voxels within each network to generate a single representative timeseries per network. Timeseries from the DMN and ECN were orthogonalized to remove shared variance between them. Using the VBA toolbox, we then fitted the following equation: $BVS_{FGAT\text{-}distant} = (\alpha DMN_{FGAT\text{-}distant}{}^{\delta} + (1\text{-}\alpha)ECN_{FGAT\text{-}distant}{}^{\delta})^{1/\delta}$

The estimated parameters from this equation are referred to as the $\alpha_{neural}$, capturing the relative contribution of DMN and ECN activities to the BVS activity, and the $\delta_{neural}$, capturing the curvature of contribution of the DMN and ECN activities to the BVS activity. We then used Pearson correlations to compare each neural parameter, estimated with the timeseries, to its respective behavioral parameter, estimated with the ratings.

Note that we performed a control analysis which takes the preprocessed fMRI data as input without regressing out task-related onsets (Supplementary Results and Figure S3).

### Statistics and reproducibility

Details of the conducted statistical analyses are reported in the subsections relative to each result in the Methods and the Supplementary Methods. Note that results were considered significant when p-values were smaller than 0.05. To ensure reproducibility, all the information regarding participants and experimental design can be found in the Methods and the Supplementary Methods and in the Data and Code availability section.

## Data availability

All data can be found at https://osf.io/4jn7p/[130] or upon request.

## Code availability

All scripts can be found at https://github.com/smorenorodriguez/CreHack_NeuroValuation or upon request.

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## Author contributions

E.V., A.L.-P. and S.M.-R. designed the study. S.M.-R. collected the data and performed all analyses. B.B. performed the preprocessing of the fMRI. A.L.-P. provided template scripts for analyses. S.M.-R., A.L.-P. and E.V. wrote the article. All authors reviewed and edited the article.

## Competing interests

The authors report no competing interests.

## Additional information

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